

RS004 Week 3 — English Worksheet

Topic: Player & Keyboard Movement · **Course:** Platformer Game · **Time:** about 45 minutes

This week a **player** appears — a rectangle you control with the **arrow keys**. You will learn `pygame.Rect` to track position, read held-down keys with `pygame.key.get_pressed()`, and stop the player from sliding off the screen.

Keep these words handy: **Rect**, **position**, `get_pressed()`, **speed**, **boundary (edge)**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
player = pygame.Rect(100, 500, 50, 50)
```

What do the four numbers mean? Where on the screen is the player?


```
keys = pygame.key.get_pressed()
if keys[pygame.K_RIGHT]:
    player.x = player.x + 5
```

The right arrow is held down. What happens to the player each frame?


```
if player.x > WIDTH - player.width:
    player.x = WIDTH - player.width
```

The player runs into the right wall. What does this code stop from happening?

2 · Rect Reading

A player is `pygame.Rect(100, 500, 50, 50)` . Fill in the blanks.

```
player.x      = ____  
player.y      = ____  
player.width  = ____  
player.right  = ____ (hint: x + width)  
player.bottom = ____ (hint: y + height)
```


3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — Arrow keys should move the player smoothly while held. Right now the player only moves on a brand-new key press, one step at a time.

```
for event in pygame.event.get():
    if event.type == pygame.KEYDOWN:
        if event.key == pygame.K_RIGHT:
            player.x = player.x + 5
```

Hint: for *held* keys we use `pygame.key.get_pressed()`, not `KEYDOWN` events.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The player should stop at the left edge. Right now it disappears off the left side.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    player.x = player.x - 5
```

Hint: after moving, check if `player.x` went below 0 and pull it back.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Both arrows should work. Right now only the right arrow does anything.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_RIGHT]:
    player.x = player.x + 5
    if keys[pygame.K_LEFT]:
        player.x = player.x - 5
```

Hint: the left check is trapped *inside* the right check. They should be separate.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Tune the Feel

Start from your working player. Change one thing at a time and note the effect.

1. Set `PLAYER_SPEED` to 2, then 10. Which feels best to you, and why?

2. Change the player's colour and size. Write what you picked.

3. **Stretch:** make the player start in the middle of the screen instead of the left. Which number did you change?

5 · Build & Show 📷

Put your player on top of last week's gradient sky. It should move left and right with the arrow keys and stop at both walls.

When it works, send a **photo or video**, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I made the player with `pygame.Rect(...)` which means ...

To move it, I used `get_pressed()` because ...

I stopped it leaving the screen by ...

The speed I chose was ____ because ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video while you move the player around. Teach it to someone new. Try to use these words:

Rect, arrow keys, speed, edge, boundary.

1. Show the player moving left and right.
2. Run into a wall and show it stops.
3. Read the boundary check out loud and explain it.
4. Show what happens when you change the speed.

Write what you will say in your video. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.